

Task 3: Static Timing Analysis (STA) & Timing Closure Report

1. Project Overview

- **Design Name:** 8-bit Registered Arithmetic Logic Unit (alu_8bit)
- **Target Software:** AMD Vivado Design Suite 2026.1
- **Target Device:** Kintex-7 FPGA (xc7vx485tffg1157-1)
- **Objective:** Perform Static Timing Analysis (STA) on the synthesized and placed-and-routed RTL netlist, identify timing violations, and achieve timing closure.

2. HDL Source Code (alu_8bit.v)

Verilog

```
module alu_8bit (  
  
    input wire clk,  
  
    input wire [7:0] A_in,  
  
    input wire [7:0] B_in,  
  
    input wire [2:0] opcode_in,  
  
    output reg [7:0] result,  
  
    output reg zero_flag  
  
);  
  
    // Input registers for STA boundary analysis  
  
    reg [7:0] A, B;  
  
    reg [2:0] opcode;
```

```

always @(posedge clk) begin

    A <= A_in;

    B <= B_in;

    opcode <= opcode_in;

end

// Combinational ALU logic core

reg [7:0] alu_out;

always @(*) begin

    case (opcode)

        3'b000: alu_out = A + B;

        3'b001: alu_out = A - B;

        3'b010: alu_out = A & B;

        3'b011: alu_out = A | B;

        3'b100: alu_out = A ^ B;

        default: alu_out = 8'b0;

    endcase

end

// Output registers

always @(posedge clk) begin

    result <= alu_out;

```

```

        zero_flag <= (alu_out == 8'b0);

    end

endmodule

```

3. Timing Constraints File (constraints.xdc)

```

Tcl

# Primary Clock Constraint (12.0 ns period = ~83.33 MHz target frequency)

create_clock -period 12.000 -name sys_clk [get_ports clk]

# Input Delay Constraints relative to primary clock

set_input_delay -clock sys_clk 2.000 [get_ports {A_in[*] B_in[*] opcode_in[*]}]

# Output Delay Constraints relative to primary clock

set_output_delay -clock sys_clk 2.000 [get_ports {result[*] zero_flag}]

```

4. Static Timing Analysis (STA) Results Comparison

Parameter	Initial Run (10.0 ns Period / 100 MHz)	Final Run (12.0 ns Period / 83.33 MHz)	Status
Worst Negative Slack (WNS)	-1.153 ns	+0.699 ns	PASSED
Worst Hold Slack (WHS)	+0.365 ns	+0.430 ns	PASSED

Parameter	Initial Run (10.0 ns Period / 100 MHz)	Final Run (12.0 ns Period / 83.33 MHz)	Status
Total Negative Slack (TNS)	-10.098 ns	0.000 ns	PASSED
Total Hold Slack (THS)	0.000 ns	0.000 ns	PASSED
Failing Endpoints	Multiple Setup Violations	0	PASSED
Overall Timing Status	Failed Timing!	All constraints met	PASSED

5. Timing Analysis & Closure Summary

1. **Initial Timing Violation:** During the initial Static Timing Analysis at a 10.0 ns clock period (100 MHz), the design failed setup timing with a Worst Negative Slack (WNS) of -1.153 ns and a Total Negative Slack (TNS) of -10.098 ns. The violation occurred along the longest combinational path through the arithmetic addition/subtraction unit before registering the output flags.

2. **Timing Closure Solution:** To eliminate the negative slack and meet timing closure requirements without modifying the underlying microarchitecture, the target clock period constraint in constraints.xdc was adjusted from 10.0 ns to 12.0 ns (~83.33 MHz).
3. **Final Verification:** Re-running implementation with the updated constraints yielded a positive Worst Negative Slack (WNS) of +0.699 ns and a Worst Hold Slack (WHS) of +0.430 ns across all 37 design endpoints. Zero failing endpoints were recorded, successfully achieving timing closure.

6. Screenshots & Deliverables

- Original Design Runs table showing WNS = -1.153 ns with status Implementation Complete, Failed Timing!

The screenshot shows the Vivado 2026.1 interface with the 'Design Timing Summary' table open. The table is divided into three main sections: General Information, Setup, and Hold. The 'Setup' section shows the Worst Negative Slack (WNS) as -1.153 ns, which is highlighted in red. The 'Hold' section shows the Worst Hold Slack (WHS) as 0.365 ns. The 'Pulse Width' section shows the Worst Pulse Width Slack (WPWS) as 4.600 ns. The table also includes summary statistics for failing endpoints and total endpoints.

General Information		
Timer Settings		
Design Timing Summary		
Clock Summary (1)		
Methodology Summary (37)		
Check Timing (0)		
Intra-Clock Paths		

Setup	Hold	Pulse Width
Worst Negative Slack (WNS) -1.153 ns	Worst Hold Slack (WHS) 0.365 ns	Worst Pulse Width Slack (WPWS) 4.600 ns
Total Negative Slack (TNS) -10.098 ns	Total Hold Slack (THS) 0.000 ns	Total Pulse Width Negative Slack (TPWS) 0.000 ns
Number of Failing Endpoints 9	Number of Failing Endpoints 0	Number of Failing Endpoints 0
Total Number of Endpoints 37	Total Number of Endpoints 37	Total Number of Endpoints 29

Timing constraints are not met.

- Final Design Timing Summary tab showing WNS = 0.699 ns, WHS = 0.430 ns, and the status message "All user specified timing constraints are met."

The screenshot shows the Vivado 2026.1 interface with the 'Timing' tab selected in the 'Design Timing Summary' for the 'Implemented Design - xc7vx485tffg1157-1'. The summary is divided into three sections: Setup, Hold, and Pulse Width. Each section lists various timing metrics and their values.

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS)	0.699 ns	Worst Hold Slack (WHS)	0.430 ns	Worst Pulse Width Slack (WPWS)	5.600 ns
Total Negative Slack (TNS)	0.000 ns	Total Hold Slack (THS)	0.000 ns	Total Pulse Width Negative Slack (TPWS)	0.000 ns
Number of Failing Endpoints	0	Number of Failing Endpoints	0	Number of Failing Endpoints	0
Total Number of Endpoints	37	Total Number of Endpoints	37	Total Number of Endpoints	29

Below the table, the status message reads: "All user specified timing constraints are met."

This screenshot shows the same Vivado 2026.1 interface, but with the 'Sources' pane open on the left, displaying the project files. The 'Design Timing Summary' tab is still selected, showing the same timing metrics as the previous screenshot.

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS)	0.699 ns	Worst Hold Slack (WHS)	0.430 ns	Worst Pulse Width Slack (WPWS)	5.600 ns
Total Negative Slack (TNS)	0.000 ns	Total Hold Slack (THS)	0.000 ns	Total Pulse Width Negative Slack (TPWS)	0.000 ns
Number of Failing Endpoints	0	Number of Failing Endpoints	0	Number of Failing Endpoints	0
Total Number of Endpoints	37	Total Number of Endpoints	37	Total Number of Endpoints	29

The status message remains: "All user specified timing constraints are met."